

week-1 day-1:

Classroom Assignment:

Linear Regression Algorithm:

Chennai House pricing:

The Trained model Calculation:

Price_Lakhs in ₹L =

$$\begin{aligned} & 8.46 \text{ ₹L} \leftarrow \text{base price (minimum)} \\ & + 0.054 \text{ ₹L per sq ft} \times \text{Built-up area} \\ & + 24.1 \text{ ₹L per point} \times \text{Locality} \\ & - 0.466 \text{ ₹L per year} \times \text{Age of building} \\ & + 21.0 \text{ ₹L if Yes} \times \text{Metro station nearby} \\ & + 14.0 \text{ ₹L if Yes} \times \text{East facing} \end{aligned}$$

a) Selected th high locality points

Predicted Amount:

TOTAL AMOUNT

356 ₹L

₹3.56 Cr

calculation:

Base price — the starting value, before any feature counts

8.46 ₹L

Built-up area

$$\begin{aligned} & 0.054 \text{ ₹L per sq ft} \times 1400 \text{ sq ft} \\ & = +76.2 \text{ ₹L} \end{aligned}$$

Locality

$$\begin{aligned} & 24.1 \text{ ₹L per point} \times 10.0 \text{ points} \\ & = +241 \text{ ₹L} \end{aligned}$$

Age of building

$$\begin{aligned} & -0.466 \text{ ₹L per year} \times 10.0 \text{ yrs} \\ & = -4.66 \text{ ₹L} \end{aligned}$$

Metro station nearby

$$\begin{aligned} & 21.0 \text{ ₹L if Yes} \times \text{Yes} \\ & = +21.0 \text{ ₹L} \end{aligned}$$

East facing

$$\begin{aligned} & 14.0 \text{ ₹L if Yes} \times \text{Yes} \\ & = +14.0 \text{ ₹L} \end{aligned}$$

TOTAL 356 ₹L (₹3.56 Cr)

b) Selected the low locality point disabling metro and east facing

calculation:

Base price — the starting value, before any feature counts

8.46 ₹L

Built-up area

$$\begin{aligned} & 0.054 \text{ ₹L per sq ft} \times 1400 \text{ sq ft} \\ & = +76.2 \text{ ₹L} \end{aligned}$$

Locality

$$\begin{aligned} & 24.1 \text{ ₹L per point} \times 0.800 \text{ points} \\ & = +19.3 \text{ ₹L} \end{aligned}$$

Age of building

$$\begin{aligned} & -0.466 \text{ ₹L per year} \times 10.0 \text{ yrs} \\ & = -4.66 \text{ ₹L} \end{aligned}$$

Metro station nearby

$$\begin{aligned} & 21.0 \text{ ₹L if Yes} \times \text{No} \\ & = +0 \text{ ₹L} \end{aligned}$$

East facing

$$\begin{aligned} & 14.0 \text{ ₹L if Yes} \times \text{No} \\ & = +0 \text{ ₹L} \end{aligned}$$

TOTAL 99.3 ₹L (₹0.99 Cr)

2) Classification Algorithm

a) Age<30 and govt satisfaction less, urban

PREDICTED PREFERENCE - Party T

Party A

$$\begin{aligned} & -10.9 \text{ pts} \leftarrow \text{base points} \\ & + 0.033 \times 29.0 \text{ (Age)} = +0.948 \\ & + 1.83 \times 2.50 \text{ (Govt. satisfaction)} = +4.57 \\ & - 0.610 \times \text{Yes (Urban voter)} = -0.610 \\ & \hline & = -6.02 \text{ pts} \end{aligned}$$

Party T

$$\begin{aligned} & 11.8 \text{ pts} \leftarrow \text{base points} \\ & - 0.118 \times 29.0 \text{ (Age)} = -3.43 \\ & - 1.71 \times 2.50 \text{ (Govt. satisfaction)} = -4.28 \\ & + 1.55 \times \text{Yes (Urban voter)} = +1.55 \\ & \hline & = 5.65 \text{ pts} \end{aligned}$$

Party C

$$\begin{aligned} & 1.12 \text{ pts} \leftarrow \text{base points} \\ & + 0.109 \times 29.0 \text{ (Age)} = +3.16 \\ & - 1.12 \times 2.50 \text{ (Govt. satisfaction)} = -2.81 \\ & - 0.887 \times \text{Yes (Urban voter)} = -0.887 \\ & \hline & = 0.584 \text{ pts} \end{aligned}$$

Undecided

$$\begin{aligned} & 1.63 \text{ pts} \leftarrow \text{base points} \\ & - 0.015 \times 29.0 \text{ (Age)} = -0.440 \\ & + 0.137 \times 2.50 \text{ (Govt. satisfaction)} = +0.343 \\ & + 0.063 \times \text{Yes (Urban voter)} = +0.063 \\ & \hline & = 1.60 \text{ pts} \end{aligned}$$

highest score wins → Party T

b) Age>50 and govt satisfaction less, non urban

Party A

$$\begin{aligned} & -10.9 \text{ pts} \leftarrow \text{base points} \\ & + 0.033 \times 58.0 \text{ (Age)} = +1.90 \\ & + 1.83 \times 2.50 \text{ (Govt. satisfaction)} = +4.57 \\ & - 0.610 \times \text{No (Urban voter)} = +0 \\ & \hline & = -4.46 \text{ pts} \end{aligned}$$

Party T

$$\begin{aligned} & 11.8 \text{ pts} \leftarrow \text{base points} \\ & - 0.118 \times 58.0 \text{ (Age)} = -6.85 \\ & - 1.71 \times 2.50 \text{ (Govt. satisfaction)} = -4.28 \\ & + 1.55 \times \text{No (Urban voter)} = +0 \\ & \hline & = 0.683 \text{ pts} \end{aligned}$$

Party C

$$1.12 \text{ pts} \leftarrow \text{base points}$$

$$\begin{aligned}
&+ 0.109 \times 58.0 \text{ (Age)} = +6.32 \\
&- 1.12 \times 2.50 \text{ (Govt. satisfaction)} = -2.81 \\
&- 0.887 \times \text{No (Urban voter)} = +0 \\
\hline
&= 4.63 \text{ pts}
\end{aligned}$$

Undecided

$$\begin{aligned}
&1.63 \text{ pts} \leftarrow \text{base points} \\
&- 0.015 \times 58.0 \text{ (Age)} = -0.880 \\
&+ 0.137 \times 2.50 \text{ (Govt. satisfaction)} = +0.343 \\
&+ 0.063 \times \text{No (Urban voter)} = +0 \\
\hline
&= 1.10 \text{ pts}
\end{aligned}$$

highest score wins → Party C

Home Assignment:

1) Classification Algorithm

a) Age < 30 and Govt satisfaction less, urban

Party A	$0.001 \div 1356 = 0.0\%$
Party T	$1348 \div 1356 = 99.4\%$
Party C	$2.84 \div 1356 = 0.2\%$
Undecided	$4.71 \div 1356 = 0.3\%$

b) Age > 50 and Govt satisfaction less, non urban

Party A	$0.004 \div 356 = 0.0\%$
Party T	$4.09 \div 356 = 1.1\%$
Party C	$349 \div 356 = 98.1\%$
Undecided	$2.56 \div 356 = 0.7\%$

2) Linear Regression

a) QA salary

Calculation:

Salary_LPA in LPA =

$$\begin{aligned}
&3.10 \text{ LPA} \leftarrow \text{base salary (minimum)} \\
&+ 0.966 \text{ LPA per year} \times \text{Experience} \\
&+ 1.34 \text{ LPA per point} \times \text{Skill rating}
\end{aligned}$$

Experience: 10 years

Skill Rating: 5.20

Base salary — the starting value, before any feature counts

3.10 LPA

Experience

$$\begin{aligned}
&0.966 \text{ LPA per year} \times 10.0 \text{ yrs} \\
&= +9.66 \text{ LPA}
\end{aligned}$$

Skill rating

$$\begin{aligned}
&1.34 \text{ LPA per point} \times 5.20 / 10 \\
&= +6.97 \text{ LPA}
\end{aligned}$$

TOTAL 19.7 LPA

b) Food Delivery Time

Distance: 13.5 km

Traffic: 4.20/10

Base time — the starting value, before any feature counts

7.56 min

Distance

$2.40 \text{ min per km} \times 13.5 \text{ km}$
 $= +32.4 \text{ min}$

Traffic

$3.16 \text{ min per point} \times 4.20 / 10$
 $= +13.3 \text{ min}$

TOTAL 53.3 min